

Massa Baali

PhD Candidate · Speech & Audio Language Models · Explainable AI

Speech scientist and PhD researcher at CMU specializing in speech language models, explainable speaker verification, and speech foundation models. Proven track record of award-winning research, commercialized speech systems, and high-impact publications. Founder and technical leader with experience spanning academia, industry research, and startups.

✉ mbaali@cs.cmu.edu | in [LinkedIn](#) | G [Google Scholar](#) | © Pittsburgh, PA

WORK EXPERIENCE

PhD Researcher Apr 2024 – Present

Carnegie Mellon University · Pittsburgh, PA

- ▶ Led 4 first-author projects: CoLMbo, CoLMbo-SV, SVeritas, CAARMA
- ▶ Produced 6 research papers in first year of PhD
- ▶ Lead weekly MLSP lab meetings; mentor multiple students
- ▶ Reviewer for ICASSP; LTI PhD admissions evaluator

Shadow Instructor & Lab Instructor Jan 2025 – Present

CMU Introduction to Deep Learning (11-785)

- ▶ Deliver recitations and lectures, mentor 200+ students on DL projects
- ▶ Recorded lectures on VAEs and Diffusion Models available on YouTube

Student Researcher Summer 2025

Adobe Research · San Jose, CA (Remote)

- ▶ Research internship with the Speech team

AI Researcher Feb 2022 – Mar 2024

MBZUAI · Abu Dhabi, UAE

- ▶ Built end-to-end check-worthiness detection for political debates
- ▶ Core contributor to JAIS: 13B-param Arabic LLM (395B tokens)

Deep Learning Researcher Apr 2021 – Feb 2022

Kanari AI · Remote

- ▶ Dialectal Arabic ASR outperforming Google, Microsoft, Amazon
- ▶ Introduced TTS in collaboration with QCRI and CMU

Founder Apr 2015 – Jan 2019

Clerk · Remote

- ▶ Built an AI-driven recruitment platform using personality and skill inference
- ▶ Accepted into Founder Institute, MIT Bootcamp; featured in major mediaoutlets

EDUCATION

Ph.D. in Language Technologies

Carnegie Mellon University · Advisor: Prof. Bhiksha Raj
2024 – 2027

M.S. in Computer Science

American University of Beirut
2019 – 2021 **FULL SCHOLARSHIP**

B.S. in Informatics Engineering (AI)

Arab International University
2015 – 2017 **FULL SCHOLARSHIP**

SKILLS & EXPERTISE

Speaker Verification	Speech LLMs
Explainable AI	PyTorch / ESPnet
ASR / TTS Systems	Deep Learning
Management	Leadership

SELECTED PUBLICATIONS

CAARMA & SVeritas
EMNLP Findings 2025

CoLMbo: Speaker Language Model
ASRU 2025

PDAF: Phonetic Debiasing Framework
SLT 2024

ESPnet-Codec: Neural Audio Codecs
SLT 2024

JAIS: Arabic 13B LLM
NLE (submitted)

DELULU: Speaker-Aware Self-Supervised Speech Foundational Model
ACL (submitted)

ACHIEVEMENTS & AWARDS

- ★ **6 papers** in 1st year of PhD
- ★ **Full scholarships** for all degrees
- ★ **Craziest Big Idea Award** SLT-CODE 2022
- ★ **Clerk AI Startup** Founder Institute, MIT, Flat6labs
- ★ **Microsoft Research** top 5% of 520 applicants

COMMUNITY & TEACHING

- ♦ **Mentor students** at CMU LTI
- ♦ **TEDx Lead Organizer** First in Country; WSJ, Guardian
- ♦ **OURCS Mentor** 48-hour demo with undergrads
- ♦ **Jusoor NGO** Taught refugees math, English, CS

LANGUAGES & INTERESTS

Arabic English 🏀 Basketball 🎾 Tennis